A Compact Semilattice Obstructs the Literal Cocommutative Group-C^* Classification

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, literal scope, for expert review. The compact semilattice $([0, 1], \min)$ refutes the direct isomorphism reading of the universal cocommutative classification suggested in Remark 3.19 of Das–Mrozinski. The source's undefined broader phrase "arise from" is not claimed to be exhausted.

1 Source statement and precise scope

The source is B. Das and C. Mrozinski, *On a quantum version of Ellis joint continuity theorem*, arXiv:1502.02469, published in *Illinois Journal of Mathematics* 60 (2016), 19–40 [1]. Their Definition 2.3 says that a compact semitopological quantum semigroup is a unital C^*-algebra A with a coassociative unital $*$-homomorphism

$$\Delta : A \longrightarrow A^{**} \mathbin{\overline{\otimes}} A^{**}$$

whose left and right slices by every functional in A^* belong to A. It is cocommutative when $\Delta = \tau\Delta$, where τ is the tensor flip.

On printed page 12, Remark 3.19 notes that cocommutative locally compact quantum groups are group C^*-algebras and suggests the corresponding classification below.

> *noncommutative compact semitopological quantum semigroup $(\mathcal{E}(\widehat{G}), \Delta')$ satisfies Assumption 1 but does not satisfy Assumption 2.*
>
> **Remark 3.19.** *The compact semitopological quantum semigroup $(\mathcal{E}(\widehat{G}), \Delta')$ appearing in Remark 3.18 (b) is cocommutative i.e. $\Delta' = \tau \circ \Delta'$, where τ is the tensor flip. It is a well-known fact (see [8, Theorem 4.2.4]) that all cocommutative locally compact quantum groups are precisely group C^*-algebras of locally compact groups. It is tempting to conjecture that all cocommutative compact semitopological quantum semigroups also arise from group C^*-algebras of locally compact groups, somewhat as in Remark 3.18 (b). However at this stage, it is not clear to us whether such characterizations are possible in general.*
>
> In the following sections, we consider situations where we can apply Theorem 3.15. Our motivations are results by Ellis in [10] and by Mukherjea and Tserpes in [14]. In particular, we provide new (C*-algebraic) proofs of these results in the compact case.

The phrase "arise from" is not defined and is qualified by "somewhat as in Remark 3.18(b)." We therefore test the most direct precise formulation:

> Every cocommutative compact semitopological quantum semigroup is isomorphic, as a C^*-bialgebra, to a group C^*-algebra $C^*(G)$ with its canonical coproduct, for some locally compact group G.

2 Proof intuition

Classical compact semigroups are already examples in the source definition: for a compact semigroup S, take $A = C(S)$ and pull functions back along its multiplication. Cocommutativity is then merely commutativity of the semigroup. Thus a compact commutative semigroup that is visibly not a group should test the proposed classification.

The operation min on the interval is the cleanest choice. It gives all slice-continuity requirements automatically because its coproduct lands in $C([0,1]^2)$. It cannot hide inside the group-C^* class: the spectrum of a commutative unital group C^*-algebra is a compact topological group, whereas $[0,1]$ is not even a homogeneous space. The coproduct also visibly fails the cancellation density enjoyed by group coproducts.

3 Counterexample and proof

Theorem 1 (Interval-min counterexample)**.** *Let $S = [0,1]$ and $m(s,t) = \min\{s,t\}$. Put $A = C(S)$ and define*

$$(\Delta f)(s,t) = f(\min\{s,t\}), \qquad f \in C(S), \quad s,t \in S. \tag{1}$$

Then (A, Δ) is a cocommutative compact semitopological quantum semigroup, but it is not isomorphic as a C^*-bialgebra to the group C^*-algebra of any locally compact group. In fact, its underlying C^*-algebra is not isomorphic to any group C^*-algebra.

Proof. The map $m : S \times S \to S$ is continuous, associative, and commutative, with identity 1. Consequently (1) is a unital $*$-homomorphism from A into $C(S \times S) \subset A^{**}\overline{\otimes}A^{**}$, associativity gives coassociativity, and commutativity gives $\Delta = \tau\Delta$.

It remains to verify the slice condition. Let $\mu \in A^* = M(S)$. For $f \in A$, a left slice has the form

$$s \longmapsto \int_S f(\min\{s,t\})\, d\mu(t), \tag{2}$$

and the right slice is identical after exchanging variables. Uniform continuity of $(s,t) \mapsto f(\min\{s,t\})$ on the compact square shows that (2) is continuous. Hence both slices lie in A, proving that (A, Δ) satisfies the source definition.

Suppose first, even without requiring compatibility of coproducts, that $C([0,1]) \cong C^*(G)$ for a locally compact group G. The group C^*- algebra is then commutative, which forces G to be abelian. Pontryagin duality gives

$$C^*(G) \cong C_0(\widehat{G}). \tag{3}$$

Because $C([0,1])$ is unital, $\widehat{G}$ is compact. Gelfand duality would therefore make $[0,1]$ homeomorphic to the compact topological group $\widehat{G}$.

No topological group is homeomorphic to $[0,1]$. A group is homogeneous: left translation sends any chosen point to any other. But an endpoint of $[0,1]$ cannot be carried to an interior point by a homeomorphism, since removing an endpoint leaves a connected space while removing an interior point disconnects it. This contradiction proves the stronger underlying C^*-algebra assertion.

For completeness, the coproduct itself also exhibits the obstruction. Every element of the algebraic span of $\Delta(A)(1 \otimes A)$ has the form

$$F(s,t) = \sum_{j=1}^{n} f_j(\min\{s,t\}) g_j(t). \tag{4}$$

On the line $t = 0$, (4) is constant in s. The same remains true after uniform closure, so this span cannot approximate the continuous function $(s,t) \mapsto s$ within error less than $1/2$. Thus Woronowicz's cancellation density fails. Canonical group coproducts in the unital case satisfy cancellation, furnishing a second bialgebra-level contradiction. □

4 Verification, limitation, and novelty

The accompanying script `code/verify_semilattice.py` checks associativity, commutativity, identity, and the cancellation obstruction in exact rational arithmetic on a 41-point grid. It reports that only the identity is invertible and that the sampled uniform cancellation obstruction is exactly $1/2$. This is a sanity check; continuity, all-point associativity, and the classification obstruction are proved above.

The conclusion is deliberately narrow. It refutes the literal universal C^*-bialgebra classification. It does not refute an unspecified claim that every cocommutative example embeds into, is compactified from, is represented inside, or is otherwise functorially associated with a group C^*-algebra. A stronger intended meaning would first need a formal definition.

A bounded search on 11 August 2026 checked the run indexes, the source TeX and PDF, the local arXiv corpus, and exact-phrase arXiv/web queries. No formal later resolution or this example was located. Details are recorded in `novelty_search.md`; novelty confidence is moderate and restricted to the precise reading above.

Human-review recommendation. Confirm that the direct-isomorphism reading is a useful interpretation of Remark 3.19. Mathematically, check Definition 2.3's slice condition and the standard implication $C^*(G)$ commutative $\Rightarrow G$ abelian; the rest is elementary.

References

[1] B. Das and C. Mrozinski, *On a quantum version of Ellis joint continuity theorem*, Illinois J. Math. 60 (2016), 19–40; arXiv:1502.02469.

[2] W. Rudin, *Fourier Analysis on Groups*, Interscience Tracts in Pure and Applied Mathematics 12, Wiley, 1962.